



Two Scripts, Both Repeating

Ontario Math C3.1 — concurrent + repeating

Name: _____

Date: _____

★ I can make two scripts repeat at the same time.

Two things at once

You can clap AND march at the same time. A computer can do that too — it can run two scripts AT THE SAME TIME. That is called concurrent.

Both scripts start on the green flag and each one has its own repeat loop. They run side by side. Watch Bit's two scripts go together.

Script A

when green flag clicked

repeat 4

move 10 steps

Script A moves Bit 4 times.

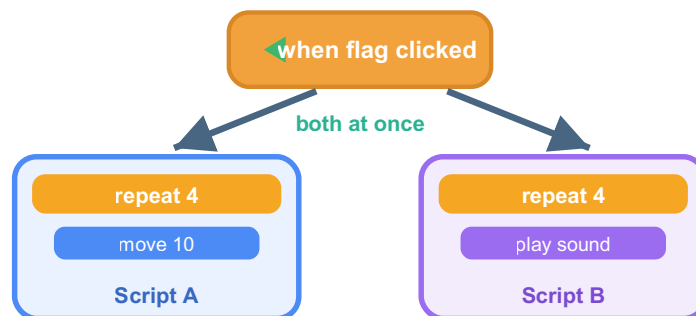
Script B (starts at the same time)

when green flag clicked

repeat 4

play sound

Script B plays a sound 4 times. Both scripts start together on the green flag — they run at the same time.



One green flag starts BOTH scripts — they loop at the same time.

1 Count it

You click the green flag once. How many times does Bit MOVE (Script A)? _____ How many SOUNDS play (Script B)? _____

2 Predict, then check

You change Script B to repeat 6 (Script A stays repeat 4). PREDICT: how many moves and how many sounds now? They run on their own — does changing B change A?

3 Write your own

Add a THIRD script so that, at the same time, Bit also spins 3 times. Fill in: when green flag clicked → repeat _____ → (turn)

4 Challenge

Both start together: Script A is repeat 5, Script B is repeat 2. Which script finishes MORE repeats, and how many more? Explain.

Big idea

Two scripts can run at the same time (concurrent). Each has its own loop and its own count — changing one does not change the other.



Two Scripts, Both Repeating

Quick facilitation notes & answer key

What this worksheet teaches

Two scripts run at the same time, each with its own repeat loop and count. Students count both, change one (the other stays the same), write a third, and compare the counts. No coding experience needed.

Before you start

No computer needed — the worksheet shows the blocks and what Bit does, so you can teach from the printout. Optional: a Scratch-style app to drag the blocks and watch Bit run — you only press the green flag.

How to lead it (about 30 minutes)

1. Show it (you do). One green flag starts both scripts; each has its own repeat number.
2. Do one together. Exercise 1 — Script A makes 4 moves, Script B makes 4 sounds.
3. Let them work. Students change one script (Ex 2), add a third (Ex 3), and compare counts (Ex 4).

Answer key (these are the verified correct answers)

Exercise 1 — 4 moves and 4 sounds. Exercise 2 — 4 moves and 6 sounds; changing Script B does not change Script A (they are separate).

Exercise 3 — repeat 3. Exercise 4 (challenge) — Script A finishes more: 5 versus 2, so 3 more repeats.

If students get stuck — and ways to adjust

Watch for: thinking changing one script changes the other. Each script counts on its own.

Make it easier: count Script A fully, then Script B.

Make it harder: ask which script finishes first if both start together.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

In plain terms: students write and run step-by-step instructions — including a "repeat" instruction — to make something happen.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

In plain terms: students read instructions, change one, and explain what the change did to the result.

You'll know it worked when

The child counts both scripts, changes one without affecting the other, and compares the counts.